



ORIGINAL ARTICLE

Accuracy in Fetal Weight Estimation by Ultrasound: A Comparative Study of Hiwale and Hadlock Methods in a Tertiary Care Hospital

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Abstract

Background Of all the methods available for fetal weight estimation, ultrasound-based estimation is the commonly used noninvasive and widely available technique. Different population-based models use a combination of fetal measurements for the estimation of fetal weight by ultrasound. Models developed for non-Indian populations give erroneous fetal weight estimates when used for Indian populations. Therefore, there is an immense need to develop an Indian-based model for sonographic fetal weight estimation. This study assesses the fetal weight from different available formulae and compares them with the actual birthweight.

Methodology This was a prospective study of 154 women assessed by ultrasound within a week of delivery. Ultrasonogram was done and head circumference, biparietal diameter, abdominal circumference and femur length were measured and estimated fetal weight was calculated using Hadlock's formula and Hiwale formula. Actual birth weight of the baby was measured after the delivery. Estimated fetal weight predicted by each formula was compared with respective neonatal actual birth weight.

Results Out of 154 pregnant women included in the study, 92.8% were of average reproductive age-group, i.e., 20–34 years with minimum age being 18 year and maximum age is 42 year. Birth weight ranged from 780 to 4200 gms, with a mean of 2619.6. Low birthweight babies constituted 34.3% ($N=53$). In our study, when weight was calculated using Hiwale method nearly 25.3% of cases had overestimated and 74.7% had underestimated the fetal weight, out of it 45.5% of cases were within the range of $\pm 10\%$ of the actual birthweight. The difference between mean estimated fetal weight and mean actual birth weight was 194.5 g.

Conclusion Hadlock's method of fetal weight estimation was found to be more accurate.

Keywords Birthweight · Fetal weight · Hadlock formula · Hiwale formula · Ultrasound estimation of fetal weight

Introduction

Fetal weight forms one of the most significant predictive parameters of neonatal outcome [1]. The inverse relationship between birth weight and infant morbidity and mortality was documented by several studies [2]. The mortality is high especially in the neonatal period and later it decreases

progressively till the age of 3 years. Two times higher perinatal and infant mortality rates are seen in these low birth infants.

The World Health Organization (WHO) based on worldwide data has recommended that neonates less than 2.5 kg be considered as low birth weight categories (LBW) and they carry relatively higher risk of morbidity and mortality with substandard growth and development in later life [3]. Nearly 80% neonatal deaths and 50% infant deaths occur in these groups. They have higher risk of developing malnutrition, recurrent infection and neurodevelopmental disorders. In the future, these low birth and growth restricted neonates are prone to develop diabetes, hypertension and coronary heart disease. Hence, low birth weight may be a key factor for adverse outcome in later life [4]

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In another end of spectrum, macrosomic fetuses which weigh more than 4 kg have higher morbidity and mortality due to the association of maternal diabetes, prolonged labor, shoulder dystocia and higher incidence of caesarian section. These babies are at risk of birth injuries like clavicle fractures, brachial plexus injuries, hypoglycemia, electrolyte imbalance and neonatal jaundice. [4]

During the last decade, estimated fetal weight has been considered as standard routine antepartum evaluation of high-risk pregnancies and deliveries. For instance, the management of diabetic pregnancy, vaginal birth after a previous cesarean section (VBAC) and intrapartum management of fetuses presenting by breech will be greatly influenced by estimated fetal weight [5,6]. Categorization of fetal weight into either small or large for gestational age may lead to timed obstetric interventions that collectively represent significant departure from routine antenatal care [6–8]. Accurate estimated fetal weight (EFW) will be helpful in planning of management, counseling on the likelihood of survival, optimal route of delivery, or level of hospital where the delivery should occur [9]. Accurate measurement of fetal weight thus forms the cornerstone in both prenatal as well as obstetric management [10]. Obstetric ultrasound has revolutionized the knowledge of fetal medicine in the present day with its diagnostic modality and precision. According to the existing literature, there is no truly accurate technique for evaluating fetal weight. Until the early 1980's, fetal weight estimation relied exclusively on clinical methods based on abdominal palpation and uterine measurements. Since the advent of ultrasound, there has been a widespread belief that ultrasound is more accurate than other methods for predicting fetal weight. Of all the different methods available for fetal weight estimation, ultrasound-based estimation is the commonly used and widely available technique [12].

The present study was aimed at estimation of fetal weight in utero by one of the formulae that have evolved over time, a recent Indian population-based formula. Earlier, ultrasound-based estimation of fetal weight used single parameters like biparietal diameter (BPD). Subsequent methods used other parameters in addition to biparietal diameter like head circumference (HC), abdominal circumference (AC) and femoral length (FL) [13], the measurements are related to the three anatomical regions—the fetal head, abdomen and the femur [14]. This has led to the development of different population-based models by using combination of fetal measurements for the estimation of fetal weight by ultrasound [15]. Fetal weight is known to be influenced by various factors such as genetic, nutritional, anthropometric, social and economic factors among different populations. Thus, a single population-based model that is equally applicable to all populations has not been developed. The accuracy of fetal weight estimation and growth monitoring depends on the ethnic specific standard model used.

Therefore, model developed for a particular population is precise for that population. Models developed for non-Indian populations give erroneous fetal weight estimates when used for Indian populations. This can indirectly lead to over or under treatment of the mother and the fetus. Therefore, to avoid such a dilemma, there is an immense need to develop an Indian-based model for sonographic fetal weight estimation [12]. Dedicated Indian nomograms are required for EFW of fetuses of specific gestational ages which will help the clinician to counsel the patient and attenders regarding the plan of action. The current study hypothesizes that the Hiwale method is more accurate than Hadlock method in estimating fetal weight in Indian population. Hence, this study was conducted.

Materials and Methods

This was a prospective study of 154 consecutive pregnant women carried out at the Department of Obstetrics and Gynecology of a tertiary care medical college hospital of south India, from September 2020 to October 2021.

Study participants included pregnant women with singleton term pregnancy admitted in the labor room for delivery or elective cesarean section and whose ultrasonogram was performed within 7 days or less of delivery, when the patient was not in labor. They had been booked for antenatal care and managed according to the existing departmental protocols. Pregnant women with obvious congenital malformations, multiple pregnancy, intrauterine fetal demise or still births were excluded from the study.

Hiwale, Mishra and Ulman (2019) study has observed that the correlation between fetal weight prediction by ultrasound and actual weight was 0.800. In the present study, expecting similar results with 80% power, 95% confidence level and considering population correlation as 0.87, the study required a minimum of 154 subjects.

Study Design

A prospective study of women with singleton pregnancy who met the inclusion criteria was included in the study. A total of 154 women with anticipated delivery within one week after considering exclusion criteria were recruited. A written informed consent was taken from all the participants. Demographic data, details of obstetric history, intrapartum events and postpartum events were recorded. Ultrasound examination was conducted by an experienced radiologist in all cases and the required fetal biometric parameters—head circumference (HC), biparietal diameter (BPD), abdominal circumference (AC) and femoral length (FL) were taken. Fetal weight prediction was made after applying the existing formulae using these parameters.

The existing formulae used for prediction of fetal weight is done by the Hadlock formula – $\text{Log}_{10}(\text{EFW}) = 1.3596 + 0.0064(\text{HC}) + 0.0424(\text{AC}) + 0.174(\text{FL}) + 0.00061(\text{BPD})$ (AC)– $0.00386(\text{AC})(\text{FL})$. The fetal weight prediction was also made by applying the new method/formula $\text{Log}_{10}(\text{EFW}) = 2.3870211110 + 0.0074323216(\text{HC}) + 0.0186555940(\text{AC}) + 0.0013463735(\text{BPD} \times \text{FL}) + 0.0004519715(\text{HC} \times \text{FL})$, brought out by Hiwale and coworkers, using the same parameters. Both the methods were compared and analyzed.

Actual birth weights of all neonates were measured using digital weighing scale immediately after birth in the labor room. The neonates in this study consisted of preterm, term and post-term neonates including the small for gestational (SGA) and large for gestational age (LGA) neonates.

The accuracy of fetal weights estimated using the existing method as well as by the method brought by Hiwale coworkers were compared with the actual birth weights. The overall performance of Hiwale method was compared with that of the existing method.

Master chart was prepared in Microsoft Excel and analyzed using the software SPSS version 2.0

Ethical approval was taken from the Institutional Ethics Committee.

Results

About 92.8% of the study population belonged to average reproductive age-group 20–34 years, 68.8% of the study population were term pregnancies. Out of 154 women, 44.2% were primigravida and 55.8% were multiparous.

Table 1 shows that amongst primiparous women, the mean of actual birthweight was comparable to mean estimated by both Hadlock and Hiwale method, p value 0.111.

However, amongst the multi, the mean of actual birthweight was inconsistent as estimated by Hadlock and Hiwale method, p value 0.012.

Table 2 depicts the mean, standard deviation (SD) and Min/Max value of the actual birth weight, Hadlock and Hiwale formula estimates. It reveals that when compared to the two methods with the actual birth weight, Hadlock (2646.7) was found to be comparable to the actual birth weight (2619.6) followed by Hiwale method (2425.1) with SD being 658.294, 621.746 and 477.507, respectively. It shows that there is a significant difference in the estimated weights between the two methods, namely Hadlock and Hiwale formula at 0.002 level of significance.

In the weight group of 1500–2499 g, both Hadlock and Hiwale method fetal weight estimation was comparable to actual birthweight with 2194.3, 2072.4 and 2063.9 g respectively and SD 375.274, 310.998 and 311.263 respectively with p value of 0.145, which is statistically not significant as shown in Table 3.

Table 4 shows Hadlock method had overestimate of 16.2% ($n = 25$), 18.8% ($n = 29$), 20.8% ($n = 32$) of < 5%, 5–10%, > 10% of actual birthweight, respectively, whereas Hiwale method had overestimate of 6.5% ($n = 10$), 3.2% ($n = 5$), 15.6% ($n = 24$) of < 5%, 5–10%, > 10% of actual birthweight. Hence, it has over all less overestimate percentage compared to Hadlock method. But, Hadlock method had less underestimate rate, i.e., 18.8% ($n = 29$), 14.3% ($n = 14.3$), 11% ($n = 17$) of < 5%, 5–10%, > 10% of actual birthweight, respectively. Hiwale method had underestimated 20.8%, 14.9% and 39% of < 5%, 5–10%, > 10% of actual birthweight, respectively.

Table 1 Pairwise comparison between different methods of measuring birth weight

		Mean diff	SE of Diff	<i>P</i> value
Primiparous	Actual versus Hadlock	– 13.3	33.687	0.695
	Actual versus Hiwale	180.6	38.905	0.186
	Hadlock versus Hiwale	193.9	20.401	0.145
Multiparous	Actual versus Hadlock	– 38.0	29.733	0.204
	Actual versus Hiwale	205.4	33.778	< 0.001
	Hadlock versus Hiwale	243.4	16.293	< 0.001

Bold value signifies the significant P values

Table 2 Comparison of Hadlock and Hiwale method with actual birth weight using ANOVA

	<i>N</i>	Mean	SD	Min. (weight in grams)	Max. (weight in grams)	<i>p</i> value
Actual weight	154	2619.6	658.294	780	4200	0.002
Hadlock formula estimate	154	2646.7	621.746	843	3740	
Hiwale formula estimate	154	2425.1	477.507	1086	3317	

Table 3 Comparison between different methods of measuring birth weight in different weight groups

Birth Weight (grams)		<i>N</i>	Mean (grams)	Standard Deviation	Min.(grams)	Max.(grams)	<i>P</i> value
< 1000	Actual	3	886.7	100.664	780	980	0.033
	Hadlock	3	994.3	132.576	843	1090	
	Hiwale	3	1225.0	121.791	1086	1313	
1000–1499	Actual	9	1355.6	95.277	1220	1480	0.014
	Hadlock	9	1437.4	142.055	1215	1663	
	Hiwale	9	1536.4	116.478	1377	1696	
1500–2499	Actual	41	2063.9	311.263	1500	2480	0.145
	Hadlock	41	2194.3	375.274	1522	2862	
	Hiwale	41	2072.4	310.998	1310	2538	
2500–3999	Actual	100	2997.4	309.991	2520	3820	<0.001
	Hadlock	100	2983.9	344.668	2159	3740	
	Hiwale	100	2680.5	276.180	1949	3317	
> =4000	Actual	1	4200.0		4200	4200	-
	Hadlock	1	3311.0		3311	3311	
	Hiwale	1	2948.0		2948	2948	

Bold value signifies the significant *P* values

Table 4 Percentagewise under and over estimation

Percentage Estimate	No. of cases with over estimation				No. of cases with under estimation			
	Hadlock		Hiwale		Hadlock		Hiwale	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
< 5	25	16.2	10	6.5	29	18.8	32	20.8
5–10	29	18.8	5	3.2	22	14.3	23	14.9
> 10	32	20.8	24	15.6	17	11.0	60	39.0

Discussion

A prospective study involves 154 women with singleton pregnancy attending the labor room. EFWs were calculated by ultrasound formulas (Hadlock and Hiwale). These values were compared with the actual neonatal weight soon after the delivery.

In our study of 154 cases, 92.8% ($n = 143$) of the cases belonged to the average reproductive age-group of 20–34 years with mean maternal age of 27.4 ± 4.3 years and 23.3 ± 4.0 years age of the patient had no correlation to the fetal weight estimation, which was similar to a study conducted by Hiwale et al. [12]

In our study, when weight was calculated using Hiwale method nearly 25.3% ($n = 39$) of cases had overestimated and 74.7% ($n = 115$) had underestimated the fetal weight, out of it 45.5% ($n = 70$) of cases were within the range of $\pm 10\%$ of the actual birth weight. The difference between mean estimated fetal weight and mean actual birth weight was 194.5 g, whereas when weight was calculated using Hadlock method nearly 55.8% ($n = 86$) of cases had

overestimated and 44.1% ($n = 68$) had underestimated the fetal weight. Out of it, 68.2% ($n = 105$) of cases were within the range of $\pm 10\%$ of the actual birth weight. The difference between mean estimated fetal weight and mean actual birth weight was—27.1 g. There was overall over-estimation in Hadlock method of fetal weight estimation for Indian population, which is similar to a study done by Yajnik CS et al. [13]. Their studies have showed that Indian fetuses have lesser birth weight and are smaller in all body measurements.

The accuracy of Hiwale method of fetal weight estimation obtained in our study was highest in the birth weight range of 1500–2499 g and lowest for the low birthweight category of 1000–1499 g, as shown in Table 3. This is in consensus with several investigators like Dudley NJ [14] who have shown that the ultrasound method is best for estimating fetal weight in the normal weight category. As Indian fetuses are smaller compared to western ones, Hiwale method seems closer to the actual weight. In low birthweight category below 1500 g, it shows a wide variation of systematic and random errors, and in high birthweight category, there is a general tendency to underestimate fetal weight.

In our study, both methods underestimated the fetal weight when birth weight was greater than 3500 g. This is comparable to the study made by Uma Thombarapu [15]. These findings were backed by the studies of Colman A Maharaj D [16], Akinolo S Shittu [17] and Niziurski Piase [18] where there was underestimation when fetal weight was > 4 kg and overestimation when less than 2.5 kg.

Although the results of our study revealed that the accuracy within 10% of actual birth weight in Hadlock method of fetal weight estimation was slightly higher than the Hiwale method (68.2% and 45.5%, respectively) and the difference of the accuracy was significant with p value < 0.001, the percentage wise estimation within 5%, 10% and 15% of actual birthweight was 35%, 68% and 78% in Hadlock method, 27.3%, 45.4% and 58.4% in Hiwale method,

respectively. This is similar with the previous studies by, Hiwale 2017 et al. who correctly estimated the actual birth weight within 5% in 25.50%, within 10% in 50.34% and within 15% in 71.81% of the cases using Hadlock method, respectively. Hiwale 2019 et al. ¹² showed that the accuracy within 10% of actual birthweight were 47.22% and 77.79% in Hadlock and Hiwale methods respectively and also Hebbar et al. showed the accuracy within 5% and 10% of actual birthweight in Hadlock methods were 45% and 78%, respectively.

The prediction error using both the formulae showed Hadlock method to be more accurate in the category of birth weight ranging from 2500 to 3999 g, whereas Hiwale method for fetal weight estimation was equally good in the other extreme weight ranges as compared to Hadlock method. However, the difference in weight estimation between the two formulae was 213.5 Grams, which is still clinically acceptable (Table 5).

Conclusion

Antenatal fetal weight can be estimated with considerable accuracy by Hiwale as well as Hadlock's formula.

Both the method of estimation of birth weight may be as accurate except in low birth weight babies. Therefore, when the Hiwale method suggests weight smaller than 1000 gms, subsequent Hadlock method of estimation is recommended to yield a better prediction and to further evaluate the fetal well-being.

Hiwale method is very accurate in predicting birth weight in 1500 to < 2499 g category than the Hadlock formula. But Hadlock prediction of birth weight is more accurate than Hiwale methods in all weight groups particularly when any growth abnormalities are present.

Among the two, Hadlock methods of fetal weight estimation was found to be more reliable in terms of showing less

Table 5 Percentagewise $\pm 10\%$ under and overestimate according to weight categories

ABW		Percentage Estimate		Total	P value
		$\pm < = 10\%$	$\pm > 10\%$		
< 1000	Hadlock	1	2	3	1.000
		33.3%	66.7%	100.0%	
	Hiwale	1	2	3	100.0%
		33.3%	66.7%	100.0%	
1000–1499	Hadlock	5	4	9	0.046
		55.6%	44.4%	100.0%	
	Hiwale	1	8	9	100.0%
		11.1%	88.9%	100.0%	
1500–2499	Hadlock	25	16	41	0.654
		61.0%	39.0%	100.0%	
	Hiwale	23	18	41	100.0%
		56.1%	43.9%	100.0%	
2500–3999	Hadlock	74	26	100	< 0.001
		74.0%	26.0%	100.0%	
	Hiwale	46	54	100	100.0%
		46.0%	54.0%	100.0%	
> = 4000	Hadlock	1	0	1	1.000
		100.0%	0.0%	100.0%	
	Hiwale	1	0	1	100.0%
		100.0%	0.0%	100.0%	

Bold value signifies the significant P values

mean error/kg of birth weight and less difference in mean estimated fetal weight from actual birth weight.

A small sample size ($n = 154$) with variable number of neonates in each weight group might have been the limiting factor for the present outcome. Hence, a larger multicentric study would lead to accurate results.

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Declarations

Conflict of interest Dr. Siri Ganesh, Dr. Jyothi GS and Dr. Poojashree K S declare that they have no conflict of interest.

Ethical Approval All procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (Ethical Review Board, MS Ramaiah Medical College, Bengaluru) and with the Helsinki Declaration of 1975, as revised in 2008(5).

Informed Consent Informed consent was obtained from all the patients for being included in the study.

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